

# Global COVID-19 Tracker & World Map

Streamlit Data Visualization Dashboard

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# Project overview

- **Goal:**

To build an interactive streamlit app hosted on Hugging Face that tracks global COVID-19 cases, deaths, and vaccinations across countries and time."

- **Tools Used:**

Streamlit, Python, Pandas, etc

- **Data Source:** CSSE COVID-19 dataset from Johns Hopkins University

# Data sources

- The dashboard uses the sourced from:
  - csse\_covid\_19\_daily\_reports folder
  - csse\_covid\_19\_time\_series folder
- These files are provided by CSSE at Johns Hopkins University
- Data coverage:
  - January 22, 2020 – March 9, 2023

```
csse_covid_19_daily_reports > 01-01-2021.csv > data
1 i,Admin2,Province_State,Country_Region,Last_Update,Lat,Long_,Confirmed,Deaths,Recovered,Active,Combined_Key,Incident_Rate,Case_Fatality_
2 fghanistan,2021-01-02 05:22:33,33.93911,67.709953,52513,2201,41727,8585,Afghanistan,134.89657830525067,4.191343095995277
3 lbania,2021-01-02 05:22:33,41.1533,20.1683,58316,1181,33634,23501,Albania,2026.409062478282,2.025173194320598
4 lgeria,2021-01-02 05:22:33,28.0339,1.6596,99897,2762,67395,29740,Algeria,227.80986075975434,2.7648477932270237
5 ndorra,2021-01-02 05:22:33,42.5063,1.5218,8117,84,7463,570,Andorra,10505.403481524623,1.0348650979425895
6 ngola,2021-01-02 05:22:33,-11.2027,17.8739,17568,405,11146,6017,Angola,53.45298103210258,2.305327868852459
7 ntigua and Barbuda,2021-01-02 05:22:33,17.0608,-61.7964,159,5,148,6,Antigua and Barbuda,162.36418593252185,3.1446540880503147
8 rgentina,2021-01-02 05:22:33,-38.4161,-63.6167,1629594,43319,1426676,159599,Argentina,3605.6333316274217,2.658269483073698
```

# Data overview

- The dataset used for this analysis report includes **countries worldwide including** Province\_State, the number of cases, deaths, Vaccinations and time series record from **2020-2023** year.

The key fields include:

| 0  | Column              | Data Format |
|----|---------------------|-------------|
| 1  | Province_State      | str         |
| 2  | Country_Region      | str         |
| 3  | Date                | datetime    |
| 4  | Confirmed           | int         |
| 5  | Deaths              | int         |
| 6  | Admin               | int         |
| 7  | Recovered           | int         |
| 8  | Lat                 | float       |
| 9  | Long                | float       |
| 10 | Case_Fatality_Ratio | float       |
| 12 | Incident_Rate       | float       |



# Data processing

## 1. Sourced from:

- **Daily case reports folder**(csse\_covid\_19\_daily\_reports folder)
- **Time series folders** (csse\_covid\_19\_time\_series folder)

## 2. Cleaning data:

- Removed missing or duplicate entries
- Remove non countries list in Country\_Region & Province\_State columns
- Filtered relevant fields (Confirmed, Deaths, Vaccinations)

# Data processing - Daily case reports folder

## 1. Collect All Daily Reports:

- Read files from the folder `csse_covid_19_daily_reports`
- Each file represents one day (e.g., 12-31-2022.csv)

## 2. Parse and Sort Dates:

- Extract the date from each filename
- Sort all files chronologically

## 3. Group by Month in each year:

- For each (year, month) combination, track the **latest available date**

## 4. Select Final Dataset

- Keep only the **last day's report of each month in each year**
- Use it to generate yearly summaries (cases, deaths, vaccinations)

# Data processing - Incident Rate & Case Fatality Ratio

## 1. Metric Calculations:

- **Case Fatality Ratio** =  $(\text{Total Deaths} / \text{Total Cases}) \times 100$

## 2. Select Relevant Columns:

- Country\_Region, Incident\_Rate, Case\_Fatality\_Ratio

## 3. Group by Country:

- Aggregate data for each country
- Calculate **mean values** across months or time periods

# Data processing – Time series folder

## 1. Load Time Series Dataset:

- Read files from the folder `csse_covid_19_time_series`
- Contains daily cumulative values across all countries

## 2. Transform to Time Series Format;

- Convert wide format to long format (Date-value pair)
- Use **melt function** to reshape the data (*e.g., from one column per day → one "Date" column*)

## 3. Filter by Country:

- Select data based on user's input (*e.g., only rows for "Germany"*)

## 4. Aggregate Values;

- Group by Date
- Sum all regional data per country to get national totals

## 5. Ensure Chronological Order:

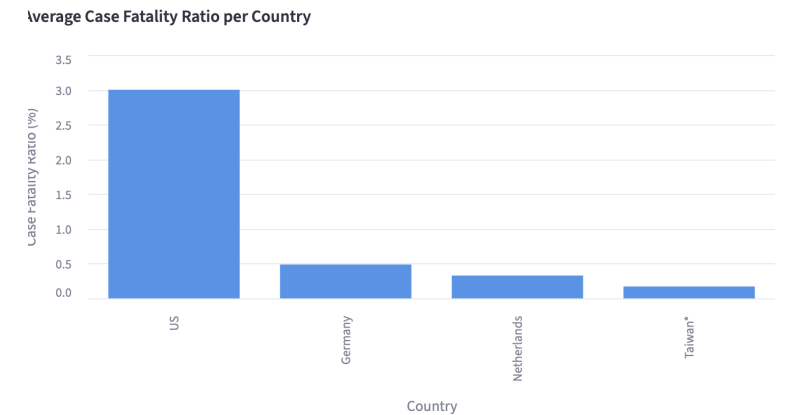
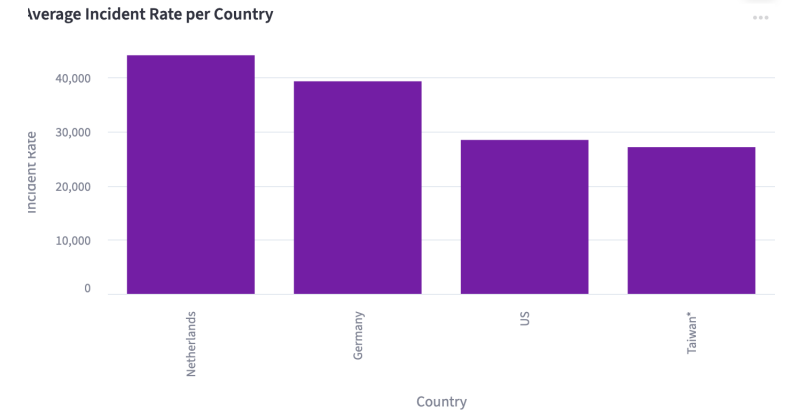
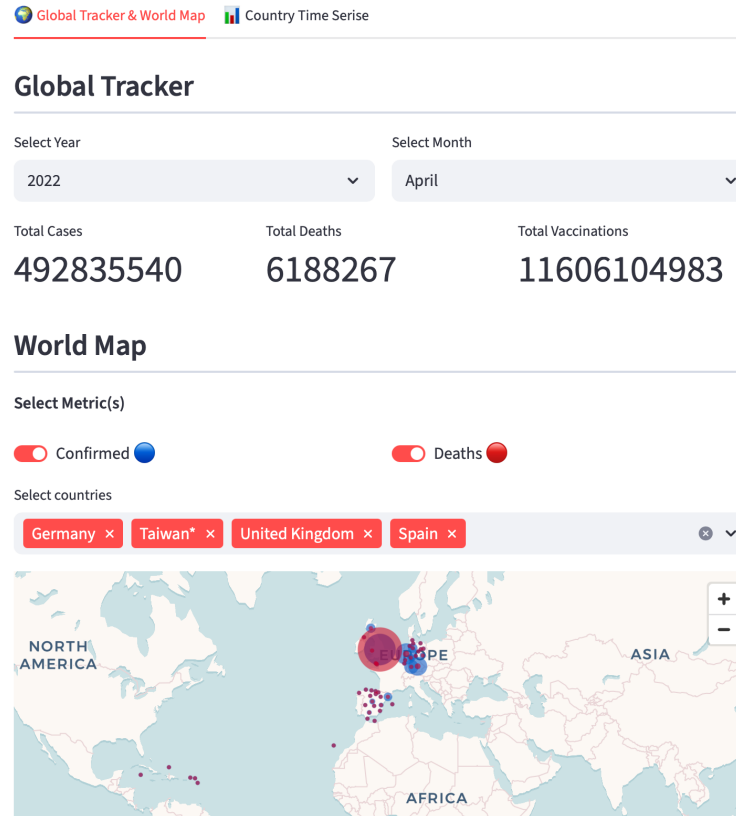
- Sort by Date to prepare for time-based plotting



# App Navigation Guide

## 1. Global Tracker & World Map

- View global statistics for a selected year and month
- Filter by metrics: Confirmed cases (blue) or Deaths (red)
- Select countries to compare on the map
- View bubble sizes by data magnitude
- Explore bar charts showing average incident rates and case fatality ratios for the countries you select



# App Navigation Guide

## 2. Country Time Series

- Choose a country to view its cumulative trend
- Line chart shows confirmed cases/deaths/vaccinations/recoveries progression over time
- Useful for identifying surges and flattening patterns



# “ Why This Dashboard Matters”

## **1. Informed Decision-Making**

- National policymakers can monitor infection rates and fatality ratios across countries and over time.
- Enables data-driven interventions (e.g., strict regulation or healthcare support where metrics are worse).

## **2. Targeted Resource Allocation**

- Highlights countries or regions with persistently high infection or mortality rates.
- Facilitates more effective distribution of vaccines, medical supplies, and healthcare support.

## **3. Historical Pattern Analysis**

- Time series charts help users understand the impact of past events (e.g., new prevention, public policies).
- Useful for future pandemic preparedness and evaluating the effectiveness of interventions.

Questions?

